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Strategy for constructing calibration sets based on a derivative spectra information space consensus

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ABSTRACT

Constructing an excellent calibration set is crucial to ensuring accurate multivariate calibration of spectra data. The purpose of this paper is to present an improved Kennard–Stone (KS) calibration set construction strategy based on different derivative spectra information spaces, termed Consensus Kennard–Stone (CKS). The core idea is to make full use of different derivative spectra information spaces when constructing the calibration set using a consensus selection method as well as to improve the prediction performance of the multivariate regression model. The experimental results from two public spectra datasets indicate that the proposed CKS strategy can use a more appropriate subset of samples for constructing the calibration set in the multivariate regression model and has superior predictive performance compared with the existing classic sample-selection KS strategies.

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1. Introduction

Multivariate regression model have been widely used in the spectra quantitative analysis and chemical analysis fields [1–4]. The most commonly used multivariate calibration technique is partial least squares (PLS) and the performance of the PLS model heavily depends on the calibration set samples. An appropriate calibration set that includes a more representative sample should be constructed from a pool of samples.

Several approaches have addressed the problem of constructing a representative subset as a calibration set for the PLS model from a large dataset [5–11]. Among them, the Kennard–Stone (KS) algorithm is well known in the field of chemometrics [5]. The KS algorithm is aimed at separating the samples in the calibration and test sets. It selects representative samples to uniformly cover the spectra data space by maximizing the Euclidean distances between the instrumental response vectors of the selected samples. However, the KS algorithm is currently only used in a single spectra information space, which is mainly in the original or smooth spectra space. As a result, the implementation of the KS algorithm relies heavily on the quality of the spectra. In fact, useful information can be thoroughly mined from the samples by combining the multiple spectra information space, especially the derivative spectra space. Therefore, the inclusion of different spectra information spaces in constructing the calibration set process often results in a more reasonable distribution of the calibration subset, improving the quality of the multivariate calibration. This topic merits further study.

An improved Kennard–Stone calibration construction strategy based on derivative spectra information space, termed Consensus Kennard–Stone (CKS), is proposed. The core idea is to fully use different spectra information spaces when constructing the calibration set via the consensus selection method as well as improve the model performance. Two different public spectra datasets are used to evaluate the proposed CKS strategy. The performances of the PLS model based on KS and the performance of PLS model based on CKS are compared in our experiment. We first introduce the method of obtaining derivative spectra information space, the classic KS algorithm and proposed CKS strategy in Section 2. Section 3 introduces the sample data sets used in the experiments and the calculation method. The results and discussion are presented in Section 4. A brief conclusion is presented in Section 5.

2. Materials and methods

2.1. The method of obtaining derivative spectra information space

The derivative spectra method is one of the most widely used approaches for data analysis because of its ability to extract more analytical information from the spectra. Additionally, this approach provides a better resolution of the spectral overlap by magnifying

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small differences between the spectral curves [12–15]. Calculating derivatives of the spectra data with the Savitzky–Golay (SG) algorithm is a landmark development. As a preliminary preprocessing step, it can effectively resolve overlapping signals, suppress unwanted spectral features and enhance signal properties [16–17]. However, it has some limitations and caveats whereby its usefulness heavily depends on proper selection of various parameters, such as the polynomial order and window size.

A new derivative spectra estimator (DSE) based on the singular perturbation technique was designed according to the description in a previously published paper [18]. The DSE was designed as follows:

$$\begin{cases} \dot{x}_1(v) = x_2(v) \\ \dot{x}_2(v) = x_3(v) \\ \dot{x}_3(v) = -\dfrac{P_3^3}{\varepsilon^3}(x_1(v) - u(v)) - \dfrac{P_3^2}{\varepsilon^2}x_2(v) - \dfrac{P_3^1}{\varepsilon}x_1(v) \\ y(v) = x_1(v) \end{cases} \tag{1}$$

$u(v)$represents the spectra signals from the instrument. $P_n^m = \frac{n!}{(n-m)!}$, $m = 1,2,3.n = 1,2,3$. x_1, x_2 and x_3 are the zero-order (smoothing), first-order and second-order derivative signals of $u(v)$, respectively. The parameter ε is the system perturbation parameter, and $\varepsilon > 0$. Currently, the parameter ε selection procession is performed by a trial-and-error method. Considering that RMSEP and RMSECV are suitable as evaluation indicators for the model performance, εcorresponding to lower RMSEP and RMSECV was selected. Additionally, there should be no significant differences between the performance of the calibration and test sets for successful modelling. That is, RMSEP and RMSECV are as close as possible. In the future, we will focus on designing a suitable optimizing function for automatic parameter selection.

2.2. Kennard–Stone (KS) algorithm

The classic KS algorithm focuses on selecting a representative subset from many samples. KS considers the instrumental response (X) in subset selection. To ensure a uniform distribution of the subset along the X data space, the KS algorithm follows a stepwise procedure in which new selections are taken in the regions of the space that are far from the samples already assigned to the subset. The closest sample to the mean of the data set can be considered the most representative one, and it is include as the first subset sample. For this purpose, the algorithm employs the Euclidean distances dx(m, n) between the x-vectors of each pair (m, n) of samples, which are calculated as

$$d_X(m,n) = \sqrt{\sum_{j=1}^{J}[X_m(j) - X_n(j)]^2} \qquad m,n \in [1,N] \tag{2}$$

Wherein, $x_m(j)$ and $x_n(j)$ are the instrumental responses at the jth wavelength or wave-number for samples m and n, respectively. j denotes the number of wavelengths or wave-numbers in the spectra. The procedure is repeated until a specified number of samples are achieved.

2.3. Consensus selection strategy based on derivative spectra information space

Construction of the calibration set is the key factor for the PLS models in quantitative analysis fields. The sample set should be as extensive and well distributed as possible. The calibration set should encompass the expected range of concentrations for the sample components. Consensus (ensemble) strategies have been widely used in multivariate regression modelling to improve the model performance [19–21]. However, this strategy rarely considers the process of constructing the calibration set. As a result, it is meaningful to explore construction of the calibration set via a consensus strategy. In this paper, to construct an excellent calibration set that preserves the data set information, an improved KS calibration construction strategy based on the derivative spectra space, CKS, is proposed. All samples are divided into the initial calibration set and initial validation set in the respective derivative spectra space using the KS method. CKS finds the common parts of three initial calibration data sets, which are generated from respective derivative spectra information

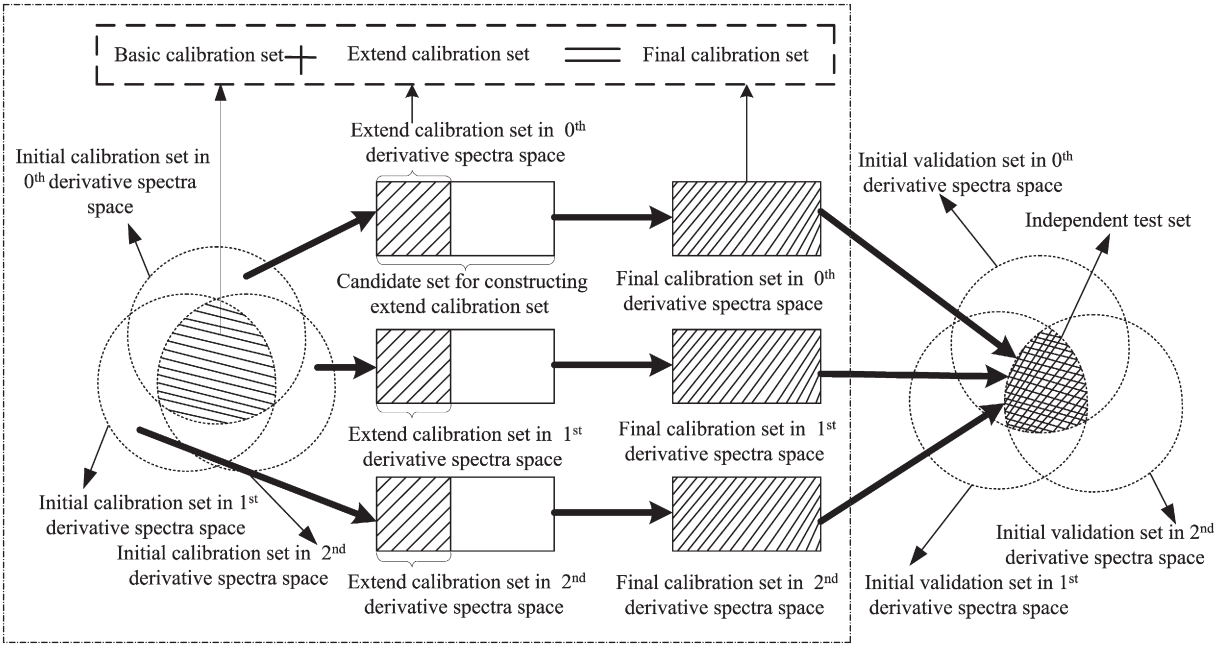


Fig. 1. Sketch map of the CKS sample selection strategy.

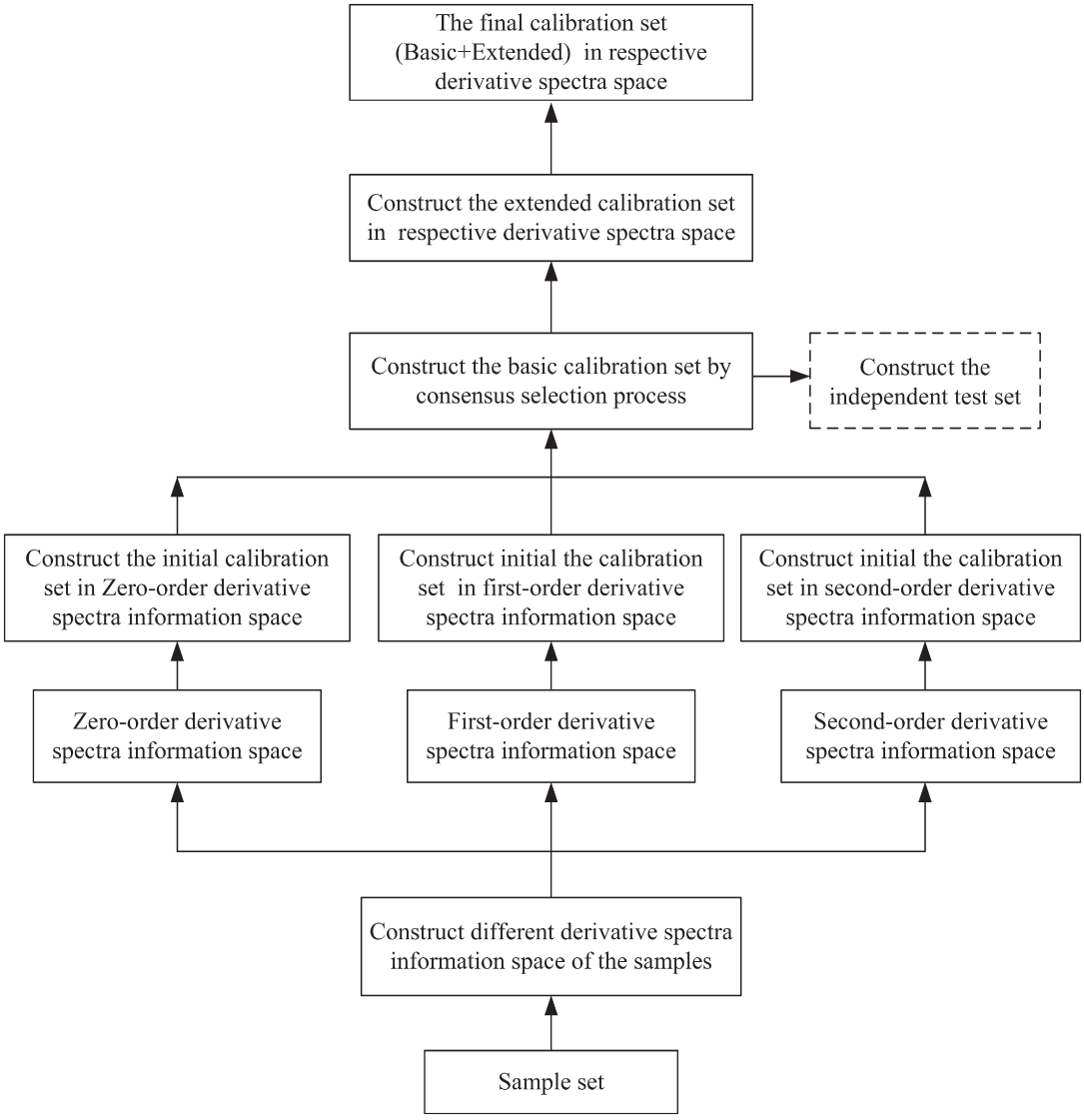


Fig. 2. The process of CKS sample selection.

space, as the basic calibration set. After constructing the basic calibration set, a candidate set, used to extend the basic calibration sets, is constructed from the remaining samples of the three initial calibration sets. Let N_{init} be the specified sample number of the initial calibration,N_{basi} be the sample number of the basic calibration set and N_{extend} be the number of samples selected from the candidate set for constructing an extended calibration set in every derivative spectra space, which is defined as

$$N_{extend} = N_{init} - N_{basic} \tag{3}$$

The sample selection principle is based on the prediction error. The relative prediction error is used as an index to determine whether the remaining sample in the initial calibration set is added to the extended calibration set. When there are no bad samples in the sample set, the sample with a larger relative predictive error is more representative and should be added to the extended calibration set. After selecting a candidate set as the validation set, PLS modelling was performed based on the basic calibration and candidate sets. The relative prediction

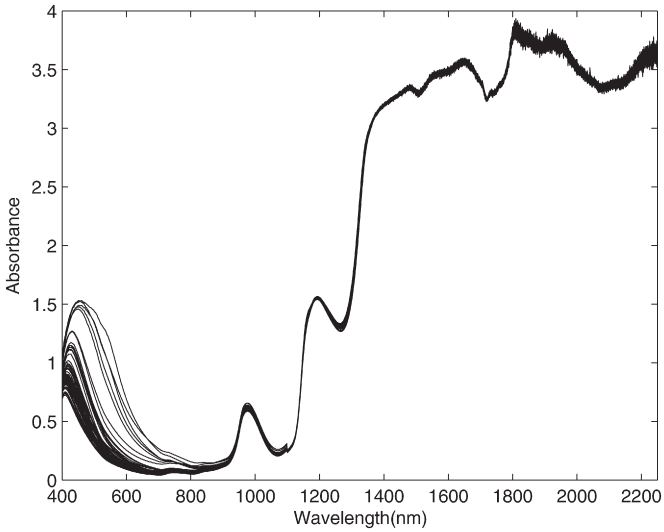


Fig. 3. Spectra of the beer sample data set.

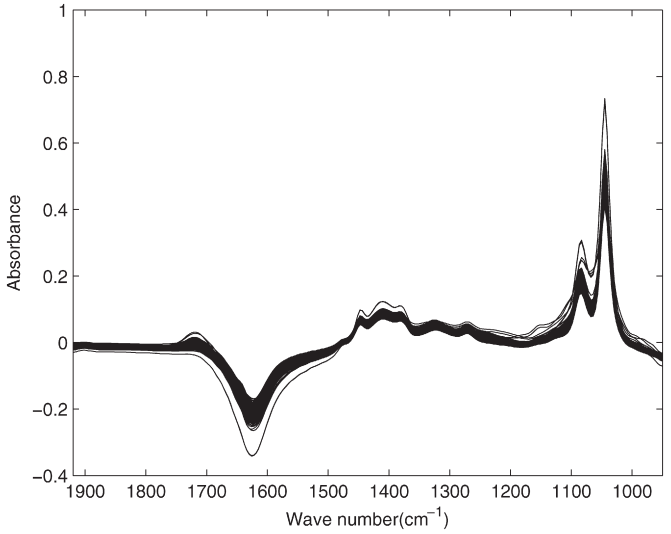


Fig. 4. Wine sample data set spectra.

error is calculated as,

$$E_{sele}(i) = \frac{Y_{rest}(i) - y_{rest}(i)}{Y_{rest}(i)} \quad (4)$$

where Y_{rest} and y_{rest} are the reference value and prediction value of samples in the candidate set. The relative error of samples is sorted in descending order. The first N_{extend} samples are selected and added to the extended calibration data set. The final calibration set consists of the basic and extended calibration sets. To verify the performance of the CKS data set partition strategy, an independent test set should be built.

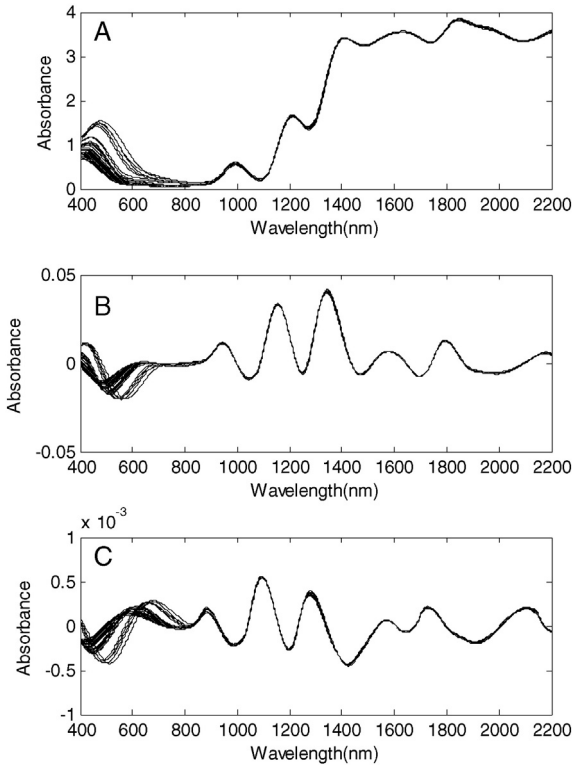


Fig. 5. Derivative spectra of beer (A) 0th derivative spectra, (B) 1st derivative spectra and (C) 2nd derivative spectra.

The common parts of three initial validation data sets are selected as the independent test set. The entire process is shown in Fig. 1.

The specific process of CKS selection is as follows:

Step 1 Construct zero-order, first-order and second-order derivative spectra information spaces of the samples.

Step 2 Select representative samples as the initial calibration and initial validation sets by the KS method in respective derivative spectra information space.

Step 3 Obtain the intersection of the three initial calibration sets as the basic calibration set by the consensus selection process.

Step 4 Select the remaining samples of three initial calibration sets as the candidate set, which is used to extend the basic calibration set.

Step 5 With the candidate set as the validation set, PLS modelling was performed based on the basic calibration and candidate sets. Obtain the relative prediction error of the samples in the candidate set.

Step 6. Sort the samples of the candidate set in descending order according to the relative error. The first N_{extend} samples are selected and added to the extended calibration data set.

Step 7. Obtain the final calibration set, which includes the basic calibration and extended calibration sets.

The entire process of CKS is also visually shown in Fig. 2.

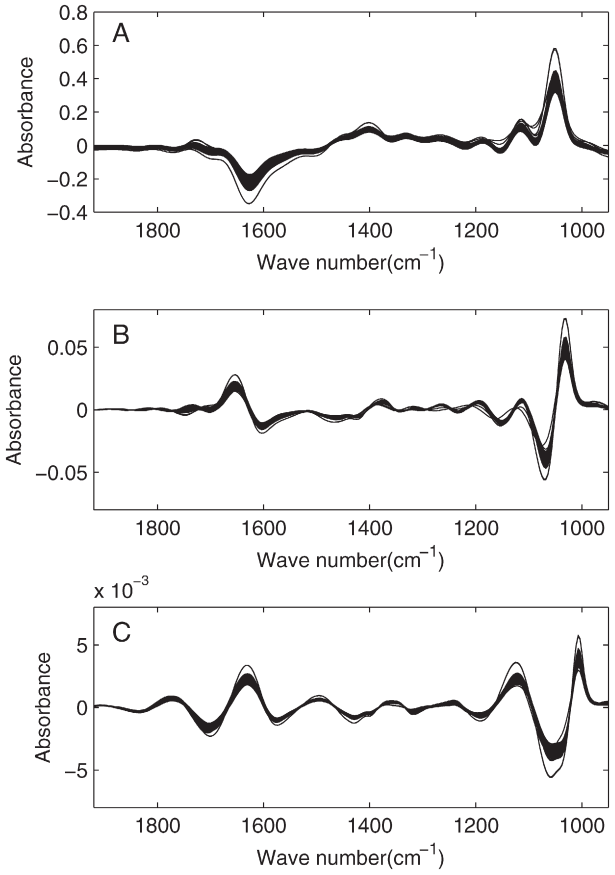


Fig. 6. Derivative spectra of wine (A) 0th derivative spectra, (B) 1st derivative spectra and (C) 2nd derivative spectra.

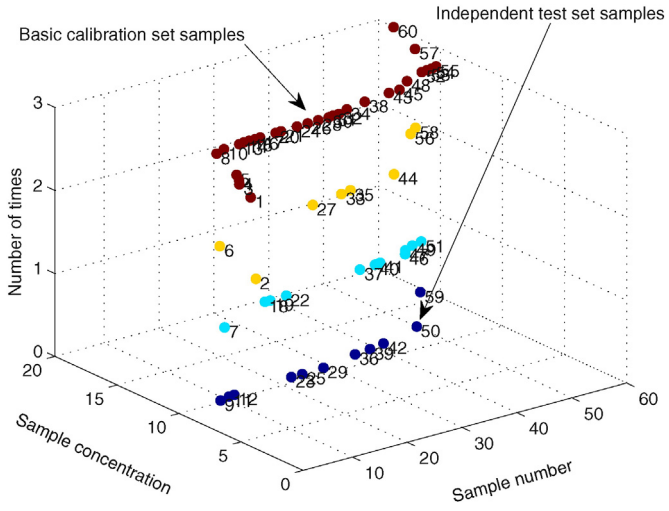


Fig. 7. Times sample selected for use in the calibration set of beer samples.

3. Experimental

3.1. Data set

The first spectra dataset is provided by Nørgaard and colleagues [22]. It consists of 60 beer samples with 926 wavelengths, which were collected from 400 to 2250 nm at an interval of 2 nm. The spectra datasets contain substantial noise due to the strong absorption. The original extract concentration was determined by Carlsberg Brewery A/S in the range of 4.23 to 18.76% plato. The original extract concentration is an important indicator in the brewing industry, which represents the ability of yeast to ferment alcohol in beer production. In the visual spectrum, from 400 to 700 nm, there is substantial variation due to variation in the visual appearance from light beers to dark beers. For the part that is greater than 1350 nm, there is substantial noise because of the strong water absorption. The raw spectra of the beer dataset are shown in Fig. 3.

The second spectra dataset was provided by Prof. Marc Meurens [23]. It contains 124 wine samples, and each spectrum consists of 256 spectral variables, which were collected from 1920 to 950 cm^{-1}. The alcohol content varies from 7.48 to 15.5% by volume. The original goal was to estimate the alcohol level in a wine sample from its mid-infrared spectrum. The No. 83 sample was abnormal and eliminated. Fig. 4 shows the wine spectra.

3.2. Calculation

The choice of the number of latent variables (Lvs) in the PLS model is a compromise between the minimum predicted residual error sum of squares (PRESS) and model complexity. The basic principle is that the model with the lowest number of latent variables is adopted on the condition that the PRESS for this model is not significantly greater than the minimum PRESS. Haaland and Thomas reported on a criterion based on testing the significance of the incremental changes in PRESS with an F-test [24]. To evaluate the PLS model performance, we used the following indicators: the root mean square error of cross-validation (RMSECV), root mean squared error of prediction (RMSEP), coefficient of determination (R^2) and residual predictive deviation (RPD). All programs are written in Matlab 7.10 (Mathworks, Inc., Natick, MA, USA) and run on a personal computer with a 2.20 GHz Intel Core 2 processor, 4 GB RAM, and Windows 7.0 operating system. Prior to building the PLS model, all data were mean-centred. No other pre-processing method was used.

3.3. Derivative spectra space construction

Construction of the high quality zero-order, first-order and second-order derivative spectra spaces is key to consensus selection. In this paper, zero-order, first-order and second-order derivative spectra spaces of the beer data set were constructed using DES, and the corresponding values of parameter ε were 0.028, 0.034 and 0.052, respectively. The beer derivative spectra are shown in Fig. 5.

At the same time, the corresponding values for the parameter ε were 0.006, 0.007 and 0.011, respectively, for the wine data set. The wine derivative spectra are shown in Fig. 6.

4. Results and discussion

4.1. Beer spectra data set results

According to the presented consensus method, the partitioning results of basic calibration set for beer data are shown in Fig. 7.

The samples that simultaneously appeared in three calibration sets with different derivative spectra space were selected for use in the basic calibration set. The samples located at the bottom of the graph, where the number of times is zero, were never selected for the calibration set and were instead selected for the independent test set. These samples were also exactly at the intersection of the initial validation sets. Prediction results for the beer data set according to the PLS models based on the KS and CKS strategies are listed in Table 1.

A lower RMSEP and RMSECV value indicates better performance. The results presented in Table 1 demonstrate that the predictive performance of the PLS-CKS models is slightly better than that of the PLS-KS models. At the same time, the RMSECV values of the PLS-CKS models are lower than the RMSECV values of the PLS-KS models. Additionally, as the main indicator of the model performance, RPD is illustrated in Fig. 8.

Usually, a relatively high RPD value indicates that the model can reliably predict the chemical composition. An RPD value greater than 5.0 is considered good for quality control and an RPD value greater than 8.0 is considered excellent for quantitative analysis fields [25]. Fig. 8 shows that the RPD values of the test and calibration sets are also slightly increased for the beer data set after partitioning data set with the CKS strategy. There is no doubt that the highest RPD of the test set was obtained with the PLS-CKS method instead of with the conventional KS strategies.

4.2. Wine spectra data set results

For the wine samples data set, the partitioning results are shown in Fig. 9.

The selection process was the same as for the beer data set. The samples located at the top of the graph, where the number of times is three, were selected as the corresponding basic calibration set in each

Table 1
Prediction results of the beer data set according to the PLS models based on the Ks and CKS calibration set construction strategies.

Method	Parameter ε	Lv	Calibration set	Independent test set
			RMSECV	RMSEP
PLS-KS(0th derivative)	0.028	9	0.282(39)	0.104(11)
PLS-KS(1st derivative)	0.034	6	0.278(39)	0.116(11)
PLS-KS(2nd derivative)	0.052	5	0.287(39)	0.112(11)
PLS-CKS(0th derivative)	0.028	9	0.249(39)	0.097(11)
PLS-CKS(1st derivative)	0.034	6	0.266(39)	0.109(11)
PLS-CKS(2nd derivative)	0.052	5	0.263(39)	0.111(11)

The values in parentheses refer to the number of samples in the corresponding sample set.

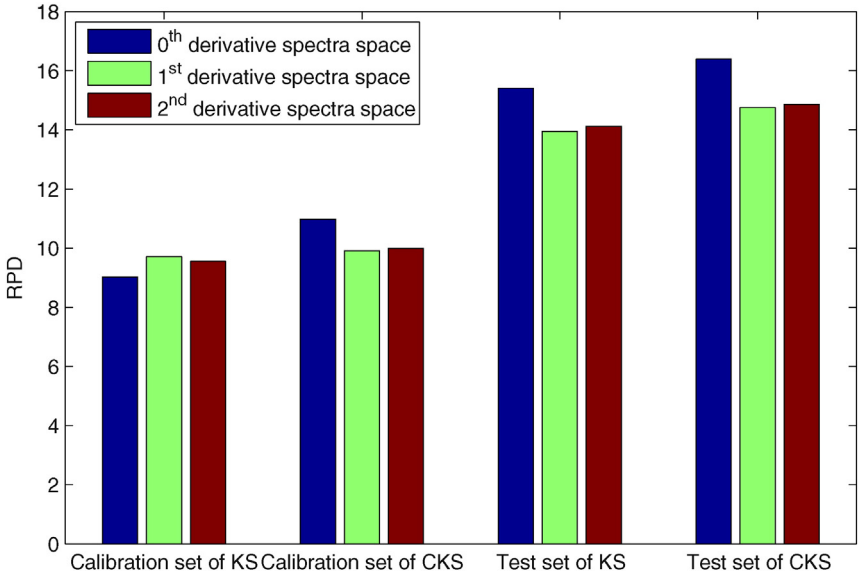


Fig. 8. RPD for different sample selection strategies for beer samples.

derivative spectra information space and the samples located at the bottom of the graph, where the number of times is zero, were selected as the independent test set. Prediction results of the wine data set by the PLS models based on the KS and CKS methods are listed in Table 2.

In the wine spectra experiment, the PLS-CKS models also have better predictive performance. At the same time, there is no obvious change in the RMSECV of the calibration set in the corresponding derivative spectra space. Specifically, among all PLS model types mentioned in Table 2, the PLS-CKS models provide lower RMSEP values. Compared with all model types, the PLS-CKS model of first-order derivative spectra provides the lowest RMSEP value. In addition, the RPD is illustrated in Fig. 10.

After selecting samples for the calibration set based on the CKS strategy, the RPD values of the PLS-CKS models are higher than the RMSECV values of the PLS-KS models in all corresponding test sets. The predictive performance of the model was significantly improved. Meanwhile, there was no obvious change in the RPD value of the calibration set in the corresponding derivative spectra space. As a result, there is no significant change in the calibration set performance. In summary, more precise or accurate prediction performance can be expected with the PLS-CKS method.

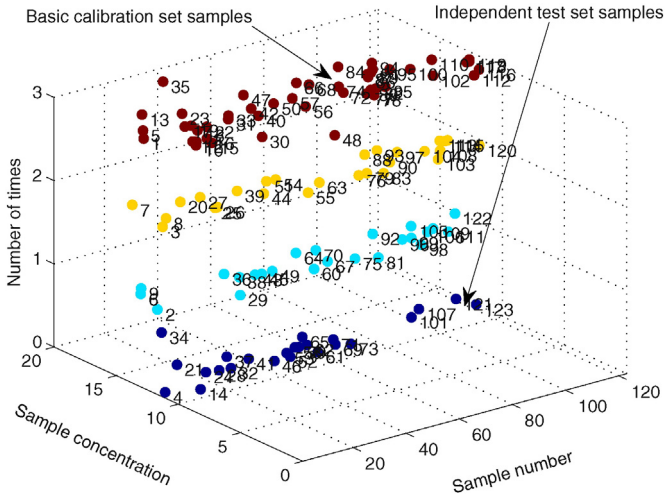


Fig. 9. Times sample selected for use in the calibration set of wine samples.

5. Conclusions

With the aim of improving the prediction accuracy and stability of the PLS model, a promising data set partitioning strategy (CKS) that takes full consideration of sample information features in different spectra spaces is proposed. In this way, the CKS can effectively utilize multiple different spectral information spaces compared with the KS algorithm mentioned above. As a result, it provides a more reasonable result for sample set partitioning in multivariate calibration and improving the prediction performance of the resulting PLS models. In addition, the CKS modelling performance seems to be affected by the sample set size and spectral signal complexity. The approach for designing sample selection rules based on the sample set characteristics remains an important topic for future work.

Conflict of interest

None.

Table 2
Prediction results of the wine data set by PLS models based on KS and CKS calibration set construction strategies.

Method	Parameter ε	Lv	Calibration set	Independent test set
			RMSECV	RMSEP
PLS-KS(0th derivative)	0.006	10	0.096(74)	0.061(24)
PLS-KS(1st derivative	0.007	9	0.097(74)	0.069(24)
PLS-KS(2nd derivative)	0.011	9	0.095(74)	0.068(24)
PLS-CKS(0th derivative)	0.006	10	0.097(74)	0.058(24)
PLS-CKS(1st derivative)	0.007	9	0.096(74)	0.053(24)
PLS-CKS(2nd derivative)	0.011	9	0.097(74)	0.059(24)

The values in parentheses refer to the number of samples in the corresponding sample set.

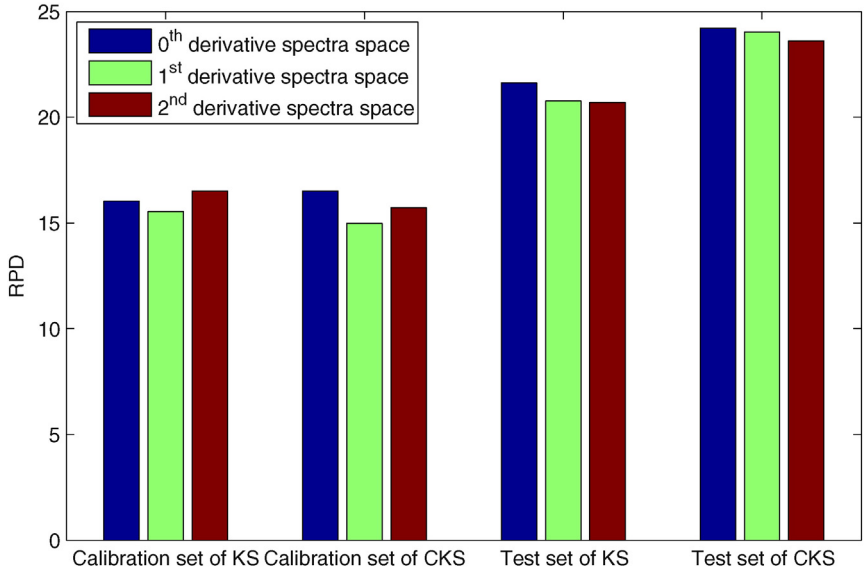


Fig. 10. RPD for different sample selection strategies for wine samples.

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